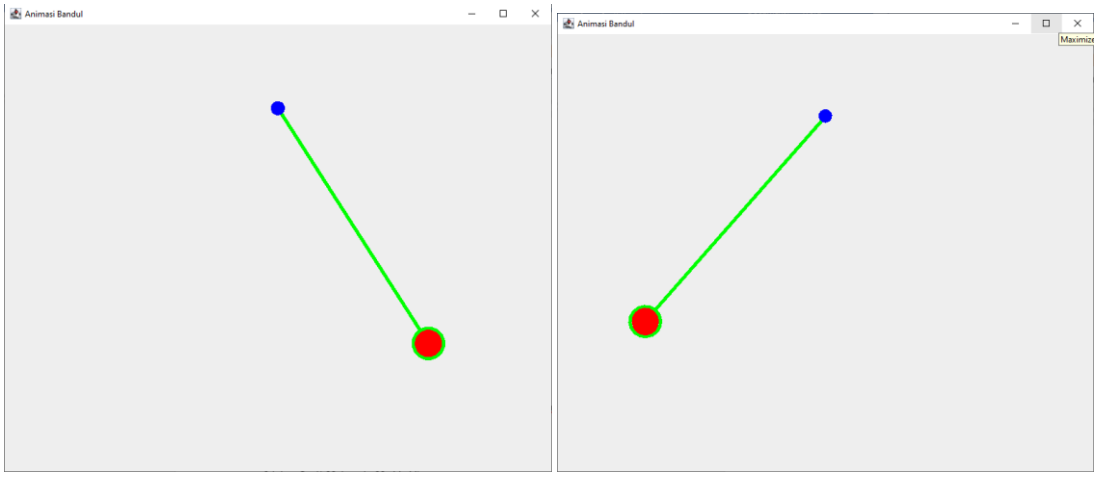
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Rangkuman 06 Apk. Komputer dan Basis Data




```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class Bandul02 extends JPanel implements ActionListener {
   private double angle = 0; // Sudut awal bandul
   private double angularVelocity = 0.03; // Kecepatan sudut
   private int length = 400; // Panjang tali bandul

   public void paintComponent(Graphics g) {
      super.paintComponent(g);
      int x = getWidth() / 2;
      int y = getHeight() / 2 -200 ;

      // Mengatur transformasi rotasi
      Graphics2D g2d = (Graphics2D) g;
      g2d.setStroke(new BasicStroke(5));
      g2d.setColor(Color.green);
      g2d.translate(x, y); // Memindahkan pusat rotasi ke titik tengah layar
      g2d.rotate(angle); // Rotasi sudut

      // Menggambar bandul
      g2d.drawLine(0, 0, 0, length); // Gambar tali bandul
      g2d.drawOval(-22, length -22, 44, 44);
      g2d.setColor(Color.red);
      g2d.fillOval(-20, length - 20, 40, 40); // Gambar bola di ujung bandul
      g2d.setColor(Color.blue);
      g2d.fillOval(-10,-10, 20, 20);
```

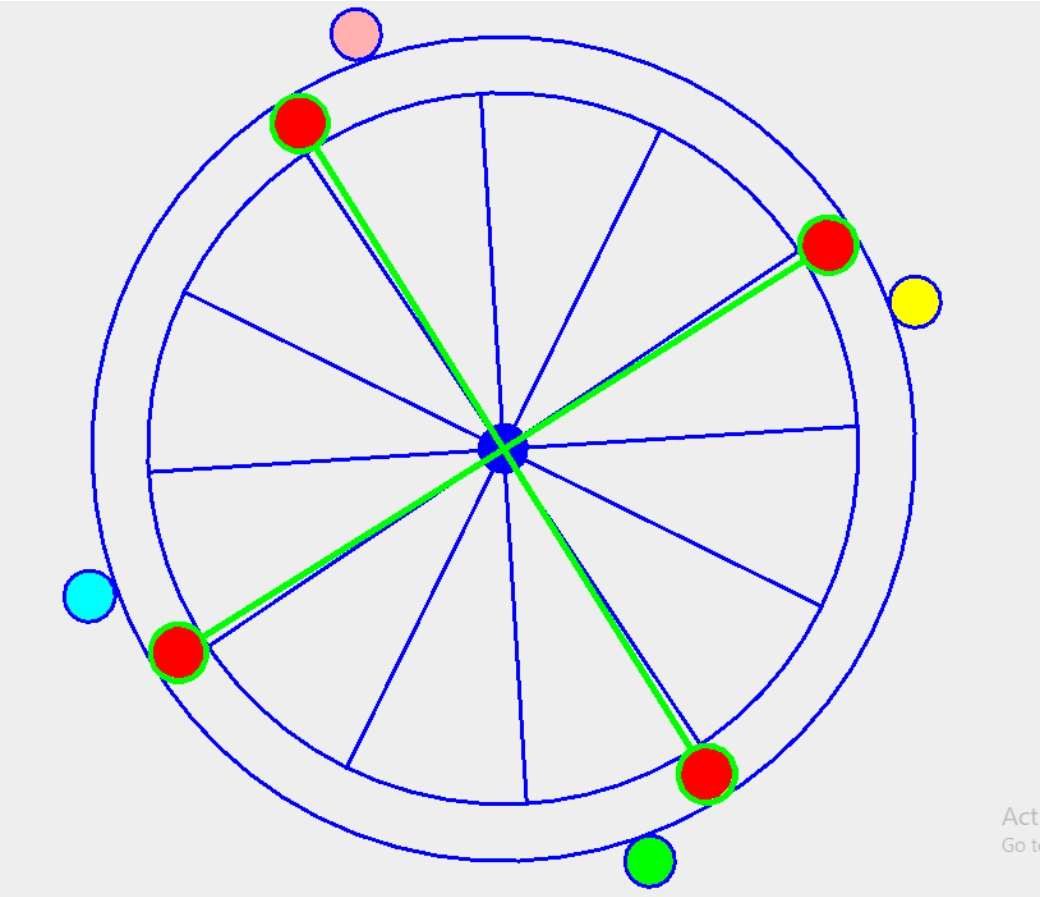
```
        // Mengembalikan transformasi ke kondisi semula
        g2d.rotate(-angle);
        g2d.translate(-x, -y);
    }

    public void actionPerformed(ActionEvent e) {
        // Mengupdate sudut bandul untuk animasi
        if(angle>=0.7 || angle<=-0.7) angularVelocity *=-1;
        angle += angularVelocity;

        repaint();
    }

    public static void main(String[] args) {
        JFrame frame = new JFrame("Animasi Bandul");
        Bandul02 bandul = new Bandul02();
        frame.add(bandul);
        frame.setSize(800, 800);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);

        Timer timer = new Timer(20, bandul); // Timer untuk mengupdate animasi setiap 20 milidetik
        timer.start();
    }
}
```



```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class Bandul03 extends JPanel implements ActionListener {
   private double angle = 0; // Sudut awal bandul
   private double angle1 = 0;
   private double angle2 = 0;
   private double angularVelocity = 0.05; // Kecepatan sudut
   private double velo = 0.02;
   private int length = 300; // Panjang tali bandul

   public void paintComponent(Graphics g) {
      super.paintComponent(g);
      int x = getWidth() / 2;
      int y = getHeight() / 2;
      // Mengatur transformasi rotasi
      Graphics2D g2d = (Graphics2D) g;
      g2d.translate(x, y); // Memindahkan pusat rotasi ke titik tengah layar
      g2d.setColor(Color.blue);
      g2d.setStroke(new BasicStroke(3));
      g2d.rotate(-angle1/2);
      g2d.drawLine(-278,0,278,0);
```

```
 for(int u=0; u<5 ; u++)
{  g2d.rotate(Math.toRadians(30),0,0);
   g2d.drawLine(-278,0,278,0);
}
//g2d.drawLine(0,-278,0,278);
g2d.rotate(Math.toRadians(-45*3),0,0);
g2d.rotate(angle1/2);
g2d.setColor(Color.blue);
g2d.fillOval(-20,-20, 40, 40);
g2d.drawOval(-322,-322,644,644);
g2d.drawOval(-278,-278,556,556);
g2d.rotate(angle); // Rotasi sudut
// Menggambar bandul
g2d.setColor(Color.green);
//g2d.fillOval(-10,-10, 20, 20);
g2d.setStroke(new BasicStroke(5));
g2d.drawLine(0, 0, 0, length); // Gambar tali bandul
g2d.drawOval(-22, length -22, 44, 44);
g2d.setColor(Color.red);
g2d.fillOval(-20, length - 20, 40, 40); // Gambar bola di ujung bandul
g2d.rotate(Math.toRadians(180),0,0);
g2d.setColor(Color.green);
g2d.drawLine(0, 0, 0, length); // Gambar tali bandul
g2d.drawOval(-22, length -22, 44, 44);
g2d.setColor(Color.red);
g2d.fillOval(-20, length - 20, 40, 40); // Gambar bola di ujung bandul
g2d.drawLine(0, 0, 0, length); // Gambar tali bandul
g2d.drawOval(-22, length -22, 44, 44);
g2d.setColor(Color.red);
g2d.fillOval(-20, length - 20, 40, 40); // Gambar bola di ujung bandul
g2d.rotate(Math.toRadians(180),0,0);
g2d.setColor(Color.green);
g2d.drawLine(0, 0, 0, length); // Gambar tali bandul
g2d.drawOval(-22, length -22, 44, 44);
g2d.setColor(Color.red);
g2d.fillOval(-20, length - 20, 40, 40); // Gambar bola di ujung bandul
g2d.rotate(Math.toRadians(-180),0,0);
g2d.setColor(Color.green);
g2d.drawLine(0, 0, 0, length); // Gambar tali bandul
g2d.drawOval(-22, length -22, 44, 44);
g2d.setColor(Color.red);
g2d.fillOval(-20, length - 20, 40, 40); // Gambar bola di ujung bandul
g2d.rotate(Math.toRadians(90),0,0);
g2d.setColor(Color.green);
g2d.drawLine(0, 0, 0, length); // Gambar tali bandul
//g2d.setColor(Color.blue);
g2d.drawOval(-22, length -22, 44, 44);
g2d.setColor(Color.red);
```

```
    g2d.fillOval(-20, length - 20, 40, 40); // Gambar bola di ujung bandul
    g2d.rotate(Math.toRadians(-180),0,0);
    g2d.setColor(Color.green);
    g2d.drawLine(0, 0, 0, length); // Gambar tali bandul
    g2d.drawOval(-22, length -22, 44, 44);
    g2d.setColor(Color.red);
    g2d.fillOval(-20, length - 20, 40, 40); // Gambar bola di ujung bandul
    g2d.rotate(Math.toRadians(180),0,0);
    g2d.setColor(Color.green);
    g2d.drawLine(0, 0, 0, length); // Gambar tali bandul
    g2d.drawOval(-22, length -22, 44, 44);
    g2d.setColor(Color.red);
    g2d.fillOval(-20, length - 20, 40, 40); // Gambar bola di ujung bandul
    g2d.rotate(Math.toRadians(-90),0,0);
    g2d.rotate(Math.toRadians(-180),0,0);
    // Mengembalikan transformasi ke kondisi semula
    g2d.rotate(-angle);
    g2d.rotate(angle2);
    g2d.setStroke(new BasicStroke(3));
    g2d.setColor(Color.yellow);
    g2d.fillOval(-19,-361,38,38);
    g2d.setColor(Color.blue);
    g2d.drawOval(-20,-362,40,40);
    g2d.setColor(Color.cyan);
    g2d.fillOval(-19,325,39,39);
    g2d.setColor(Color.blue);
    g2d.drawOval(-20,324,40,40);
    g2d.rotate(Math.toRadians(90),0,0);
    g2d.setStroke(new BasicStroke(3));
    g2d.setColor(Color.green);
    g2d.fillOval(-19,-361,38,38);
    g2d.setColor(Color.blue);
    g2d.drawOval(-20,-362,40,40);
    g2d.setColor(Color.pink);
    g2d.fillOval(-19,325,39,39);
    g2d.setColor(Color.blue);
    g2d.drawOval(-20,324,40,40);
    g2d.rotate(Math.toRadians(-90),0,0);
    g2d.rotate(-angle2);
    g2d.translate(-x, -y);
}

public void actionPerformed(ActionEvent e) {
    // Mengupdate sudut bandul untuk animasi
    if(angle>=0.8 || angle<=-1.6) angularVelocity *=-1;
    angle += angularVelocity;
    angle1 += velo;
    angle2 += velo/3.6;
```

```
        repaint();
    }

    public static void main(String[] args) {
        JFrame frame = new JFrame("Animasi Bandul");
        Bandul03 bandul = new Bandul03();
        frame.add(bandul);
        frame.setSize(800, 800);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);

        Timer timer = new Timer(10, bandul); // Timer untuk mengupdate animasi setiap 20 milidetik
        timer.start();
    }
}
```